

第二次:

② L_x, L_y, L_z 的对易关系的证明

利用定义 $L_k = \epsilon_{ijk} x_i p_j$ 展开证明 $[L_i, L_j] = i\hbar \epsilon_{ijk} L_k$ ($L_1 = L_x, L_2 = L_y, L_3 = L_z$)

$$\epsilon_{ijk} = \begin{cases} +1, & \text{偶排列} \\ -1, & \text{奇排列} \\ 0, & \text{其它} \end{cases}$$

$$\delta_{ij} = \begin{cases} 1, & i=j \\ 0, & i \neq j \end{cases}$$

Einstein 求和约定, 重复指标求和

$$\epsilon_{ijk} \epsilon_{mnk} = (\delta_{im} \delta_{jn} - \delta_{in} \delta_{jm})$$

i, j, k, l, m, n 取 $1, 2, 3$

$$[L_i, L_j] = \epsilon_{ikl} \epsilon_{jmn} [x_k p_l, x_m p_n]$$

$$= \epsilon_{ikl} \epsilon_{jmn} (x_m [x_k, p_n] p_l + x_k [p_l, x_m] p_n)$$

$$= i\hbar \epsilon_{ikl} \epsilon_{jmn} (x_m p_l \delta_{kn} + x_k p_n \delta_{lm})$$

$$= i\hbar (\epsilon_{ikl} \epsilon_{jmn} x_m p_l + \epsilon_{ikl} \epsilon_{jln} x_k p_n)$$

$$= i\hbar [(\delta_{im} \delta_{lj} - \delta_{ij} \delta_{lm}) x_m p_l + (\delta_{in} \delta_{jk} - \delta_{ij} \delta_{kn}) x_k p_n]$$

$$= i\hbar (x_i p_j - x_j p_i) = i\hbar \epsilon_{ijk} L_k$$

$$[\epsilon_{ijk} L_k = \epsilon_{ijk} \epsilon_{mnk} x_m p_n = x_i p_j - x_j p_i]$$

③ L^2 与 L_x, L_y, L_z 的对易关系证明

$$[L_i, L^2] = [L_i, L_j L_j] = L_j [L_i, L_j] + [L_i, L_j] L_j = i\hbar (\epsilon_{ijk} L_j L_k + \epsilon_{ijk} L_k L_j) = 0$$

$L_{\pm} = L_x \pm iL_y$; L_+ 与 L_- 的对易关系.

$$[L_+, L_-] = [L_x + iL_y, L_x - iL_y] = -i[L_x, L_y] + i[L_y, L_x] = -2i \cdot i\hbar L_z = 2\hbar L_z$$

L^2 与 $L_{\pm}$ 的对易关系.

$$[L^2, L_{\pm}] = [L^2, L_x \pm iL_y] = [L^2, L_x] \pm i[L^2, L_y] = 0$$

$$[\alpha, \beta] = 1, \text{ 证明 } [\alpha, \beta^n] = n \beta^{n-1}$$

数学归纳: $n=1$ 成立; 若 $[\alpha, \beta^k] = k \beta^{k-1}$ 成立;

$$\text{则 } [\alpha, \beta^{k+1}] = [\alpha, \beta^k] \beta + \beta^k [\alpha, \beta] = k \beta^{k-1} \beta + \beta^k = (k+1) \beta^k \text{ 成立.}$$

对任意 $n \in \mathbb{N}^+$, $[\alpha, \beta^n] = n \beta^{n-1}$ 成立.

$$\int \phi^* (A+B)^\dagger \psi d^3x = \int [(A+B)\phi^*]^\dagger \psi d^3x = \int (A\phi^*)^\dagger \psi d^3x + \int (B\phi^*)^\dagger \psi d^3x = \int \phi^* A^\dagger \psi d^3x + \int \phi^* B^\dagger \psi d^3x \\ = \int \phi^* (A^\dagger + B^\dagger) \psi d^3x$$

$$\int \phi^* (AB)^\dagger \psi d^3x = \int (AB\phi^*)^\dagger \psi d^3x = \int (B\phi^*)^\dagger (A^\dagger \psi) d^3x = \int \phi^* B^\dagger (A^\dagger \psi) d^3x = \int \phi^* (B^\dagger A^\dagger \psi) d^3x$$

$$\text{即 } (A+B)^\dagger = A^\dagger + B^\dagger; (AB)^\dagger = B^\dagger A^\dagger$$

若 A, B 厄米; $(A+B)^\dagger = A^\dagger + B^\dagger = A+B$, 则 $A+B$ 厄米

若 A, B 厄米, 且 $[A, B] = 0$; $(AB)^\dagger = B^\dagger A^\dagger = BA = AB$; 则 AB 厄米.

$$\hat{F} = e^{ikx} i \frac{d}{dx} = \frac{1}{\hbar} e^{ikx} p_x; \hat{G} = e^{ikx}$$

$$\text{由 } \begin{cases} [f(x), p_x] = i\hbar \frac{\partial f}{\partial x} \\ [x, g(p_x)] = i\hbar \frac{\partial g}{\partial p_x} \end{cases} \quad (\text{对 } f(x), g(p_x) \text{ Taylor 展开证明})$$

$$[\hat{F}, \hat{G}] = -[e^{ikx}, e^{ikx} \frac{p_x}{i\hbar}] = -e^{ikx} [e^{ikx}, \frac{p_x}{\hbar}] - [e^{ikx}, e^{ikx}] \frac{p_x}{\hbar} = \frac{-1}{\hbar} e^{ikx} (i\hbar) \frac{d}{dx} e^{ikx} \\ = k e^{2ikx}$$

$$\text{或 } [\hat{F}, \hat{G}] \psi = k e^{2ikx} \psi \Rightarrow \text{由 } \psi \text{ 任意性; } [\hat{F}, \hat{G}] = k e^{2ikx}$$

对 $\psi_p(\vec{r}) = C e^{i\frac{\vec{p} \cdot \vec{r}}{\hbar}}$ 进行 delta 归一

$$|C|^2 \int d^3\vec{r} \psi_p^* \psi_p = |C|^2 \int d^3\vec{r} e^{-i\frac{\vec{p} \cdot \vec{r}}{\hbar}} e^{i\frac{\vec{p} \cdot \vec{r}}{\hbar}} \\ = |C|^2 \left(\int dx e^{i\frac{(p_x - p'_x)}{\hbar} x} \right) \left(\int dy e^{i\frac{(p_y - p'_y)}{\hbar} y} \right) \left(\int dz e^{i\frac{(p_z - p'_z)}{\hbar} z} \right) \\ = |C|^2 (2\pi)^3 \delta^3\left(\frac{\vec{p}}{\hbar}\right) \uparrow = |C|^2 (2\pi\hbar)^3 \delta^3(\vec{p})$$

$$\Rightarrow |C| = (2\pi\hbar)^{-\frac{3}{2}} \quad \boxed{\delta(ax) = \frac{1}{|a|} \delta(x)}$$

$$\psi(x) = \left(\frac{1}{2\lambda a^2}\right)^{\frac{1}{4}} e^{-\frac{x^2}{4a^2}}$$

$$\begin{aligned}\psi(p) &= \int dx \psi_p^*(x) \psi(x) = \int dx \frac{1}{(2\pi\hbar)^{\frac{1}{2}}} \left(\frac{1}{2\lambda a^2}\right)^{\frac{1}{4}} e^{-\frac{x^2}{4a^2} - i\frac{px}{\hbar}} \\ &= \frac{1}{(2\pi\hbar)^{\frac{1}{2}}} \left(\frac{1}{2\lambda a^2}\right)^{\frac{1}{4}} \int dx e^{-\frac{1}{4a^2}(x + 2\frac{i p a^2}{\hbar})^2 - \frac{p^2 a^2}{\hbar^2}} \\ &= \frac{1}{(2\pi\hbar)^{\frac{1}{2}}} \left(\frac{1}{2\lambda a^2}\right)^{\frac{1}{4}} (4\lambda a^2)^{\frac{1}{2}} e^{-\frac{p^2 a^2}{\hbar^2}} = \left(\frac{2a^2}{\lambda\hbar^2}\right)^{\frac{1}{4}} e^{-p^2 a^2/\hbar^2}\end{aligned}$$

$$P(p \rightarrow p+dp) = |\psi(p)|^2 dp = \sqrt{\frac{2a^2}{\lambda\hbar^2}} e^{-\frac{2p^2 a^2}{\hbar^2}}$$

$$(\hat{p} - iC\hat{x})\psi = \lambda\psi \Rightarrow \frac{d}{dx}\psi = \left(\frac{i}{\hbar}\lambda - \frac{C}{\hbar}\right)\psi$$

$$\text{Let } \psi(x) = A e^{ax+bx^2}$$

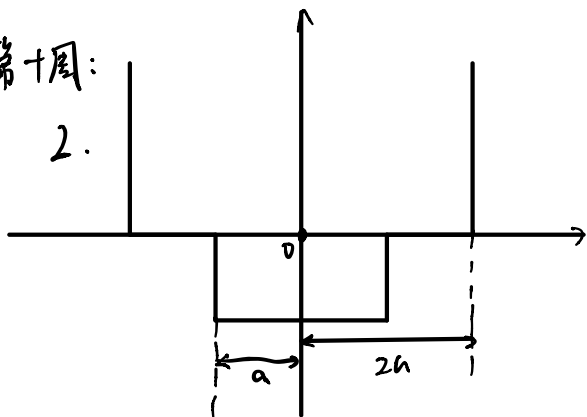
$$\frac{d}{dx}\psi(x) = (a+2bx)\psi(x) \Rightarrow \begin{cases} a = \frac{i}{\hbar}\lambda \\ b = -\frac{C}{2\hbar} \end{cases} \Rightarrow \psi(x) = A e^{\frac{i}{\hbar}\lambda x - \frac{C}{2\hbar}x^2}$$

$$\int \psi^* \psi dx = 1 \Rightarrow |A|^2 \int_{-\infty}^{\infty} e^{-\frac{C}{\hbar}x^2} dx = |A|^2 \sqrt{\frac{\lambda\hbar}{C}} = 1 \Rightarrow \psi(x) = \left(\frac{C}{\lambda\hbar}\right)^{\frac{1}{4}} e^{-\frac{C}{2\hbar}x^2} e^{\frac{i}{\hbar}\lambda x}$$

第三次.

第十周:

2.



$$1) -V_0 < E < 0; \quad \frac{d^2}{dx^2} \psi = \frac{2m(V(x)-E)}{\hbar^2} \psi = \begin{cases} k^2 \psi, & a < |x| \leq 2a, & k = \sqrt{\frac{-2mE}{\hbar^2}} \\ -k^2 \psi, & |x| \leq a, & k = \sqrt{\frac{2m(E-V_0)}{\hbar^2}} \\ \infty, & |x| > 2a, \end{cases}$$

$$\textcircled{1} \text{奇宇称: } \psi(x) = \begin{cases} A \sin kx, & |x| \leq a \\ Be^{kx} + Ce^{-kx}, & a < x \leq 2a \\ Ce^{kx} + Be^{-kx}, & -2a \leq x < -a \\ 0, & |x| > 2a \end{cases}$$

$$\text{由连续性} \left\{ \begin{aligned} \psi(a) &\Rightarrow A \sin ka = Be^{ka} + Ce^{-ka} \\ \psi'(a) &\Rightarrow Ak \cos ka = k(Be^{ka} - Ce^{-ka}) \\ \psi(2a) &= 0 \Rightarrow Be^{2ka} + Ce^{-2ka} = 0 \end{aligned} \right.$$

$$\Rightarrow \frac{\tan ka}{k} = \frac{e^{ka} - e^{3ka}}{k(e^{ka} + e^{3ka})} \Rightarrow \boxed{\frac{\tan ka}{\tanh ka} = \frac{k}{K}}$$

\textcircled{2} 偶宇称:

$$2) E > 0, \quad \frac{d^2}{dx^2} \psi = \frac{2m(V(x)-E)}{\hbar^2} \psi = \begin{cases} -k_1^2 \psi, & a < |x| \leq 2a, & k_1 = \sqrt{\frac{2mE}{\hbar^2}} \\ -k_2^2 \psi, & |x| \leq a, & k_2 = \sqrt{\frac{2m(E-V_0)}{\hbar^2}} \end{cases}$$

$$\textcircled{1} \text{奇宇称: } \psi(x) = \begin{cases} A \sin k_1 x, & |x| \leq a \\ Be^{ik_2 x} + Ce^{-ik_2 x}, & a < x \leq 2a \\ Ce^{ik_2 x} + Be^{-ik_2 x}, & -2a \leq x < -a \\ 0, & |x| > 2a \end{cases}$$

$$\text{由连续性} \left\{ \begin{aligned} \psi(a) &\Rightarrow A \sin k_1 a = Be^{ik_2 a} + Ce^{-ik_2 a} \\ \psi'(a) &\Rightarrow Ak_1 \cos k_1 a = ik_2(Be^{ik_2 a} - Ce^{-ik_2 a}) \\ \psi(2a) &= 0 \Rightarrow Be^{2ik_2 a} + Ce^{-2ik_2 a} = 0 \end{aligned} \right.$$

$$\Rightarrow \frac{\tan k_1 a}{k_1} = \frac{e^{ik_2 a} - e^{3ik_2 a}}{ik_2(e^{ik_2 a} + e^{3ik_2 a})} \Rightarrow \boxed{\frac{\tan k_1 a}{\tan k_2 a} = \frac{k_1}{k_2}}$$